



3.3.6 Wrap-Up: Ticket Out the Door

1. Use the laws of exponents to evaluate $\frac{0.2^3}{2^3} \cdot 0.1^2$. Show your work.

2. Select all of the expressions equivalent to $\frac{0.2^3}{2^3} \cdot 0.1^2$.

- ☐ 0.1^5
- ☐ $0.2^0 \cdot 0.1^2$
- ☐ $0.1^3 \cdot 0.1^2$
- ☐ $\frac{0.2^5}{0.2^3}$
- ☐ $\left(\frac{0.2}{2}\right)^3 \cdot 0.1^2$
- ☐ 0.2^5



Unit 3, Lesson 4: Negative Exponent Law



Benchmark

MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.



Learning Targets

- ☐ I can apply the laws of exponents to evaluate numerical expressions with integer exponents.
- ☐ I can generate equivalent numerical expressions using the laws of exponents for expressions with integer exponents.



3.4.1 Warm-Up: Day at Work With a Fish Biologist

Moises, an ichthyologist, or fish biologist, says, "Try to be in touch with a system that you really want to study, with a system that really drives you crazy, because that's how you nurture passion and passion is one of the most important things in becoming a researcher."

1. What is something that you are passionate about?
2. Why do you think he says that passion is one of the most important things in becoming a researcher?



3.4.2 Collaboration: Laws of Exponents

1. Complete the rule for each exponent law shown.

Exponent Law	Numerical Example	Algebraic Rule
Product of Powers	$2^3 \cdot 2^1 = 2^4$	$a^m \cdot a^n = \underline{\hspace{2cm}}$
Power of a Product	$(-7 \cdot 3)^2 = (-7)^2 \cdot 3^2$	$(ab)^m = \underline{\hspace{2cm}}$
Power of a Power	$(6^3)^4 = 6^{12}$	$(a^m)^n = \underline{\hspace{2cm}}$
Quotient of Powers	$\frac{(-3)^7}{(-3)^2} = (-3)^5$	$\frac{a^m}{a^n} = \underline{\hspace{2cm}}$
Power of a Quotient	$\left(\frac{13}{5}\right)^3 = \frac{13^3}{5^3}$	$\left(\frac{a}{b}\right)^m = \underline{\hspace{2cm}}$
Identity Exponent	$15^1 = 15$	$a^1 = \underline{\hspace{2cm}}$
Zero Exponent	$127^0 = 1$	$a^0 = \underline{\hspace{2cm}}$

2. Write an equivalent exponential expression by applying the laws of exponents.

a. $\frac{11^{14}}{11^8} = \underline{\hspace{2cm}}$

b. $10^4 \cdot 10^3 = \underline{\hspace{2cm}}$

c. $5^2 \cdot 5^{10} = \underline{\hspace{2cm}}$

d. $(4 \cdot -5)^2 = \underline{\hspace{2cm}}$

e. $(2^3)^8 = \underline{\hspace{2cm}}$

f. $\left(\frac{21}{-4}\right)^5 = \underline{\hspace{2cm}}$





3.4.3 Exploration: Finding Patterns to Evaluate Numeric Expressions With Exponents

Work with your partner to complete the following.

1. Complete the table.

Exponent Expression	Value
2^4	16
2^3	
2^2	
2^1	
2^0	

2. Describe any mathematical pattern(s) you notice regarding how the values of the expressions change as the exponent changes.

3. Complete the statement.

As the exponent ☐ increases
☐ decreases by 1, the value of the

expression can be found by ☐ multiplying
☐ dividing the

previous value by ☐ 2.
☐ 4.
☐ 8.



4. Continue the pattern to complete the table.

Exponent Expression	Value
2^1	2
2^0	1
2^{-1}	
2^{-2}	
2^{-3}	

5. Compare the values of 2^3 and 2^{-3} . What do you notice?
6. Compare the values of 2^1 and 2^{-1} . What do you notice?
7. With your partner, discuss how you can find the value of any number raised to a negative exponent. Summarize your discussion.
8. Based on the discussion with your partner, what is the value of 2^{-4} ?



3.4.4 Bring It Together: Negative Exponents



Expressions with **negative exponents** can be expressed with positive exponents: $x^{-a} = \frac{1}{x^a}$.

1. Complete the statement.

A base with a negative exponent can be rewritten using

the ☐ reciprocal
☐ opposite of the base with the

☐ reciprocal
☐ opposite exponent.

2. Complete the table.

Exponent Expression	Repeated Multiplication
a^3	$a \cdot a \cdot a$
a^2	$a \cdot a$
a^1	a
a^0	1
a^{-1}	$\frac{1}{a} = $
a^{-2}	$\frac{1}{a} \cdot \frac{1}{a} = $
a^{-3}	$\frac{1}{a} \cdot \frac{1}{a} \cdot \frac{1}{a} = $



Negative Exponent Law

$$a^{-1} = \frac{1}{a} \quad \text{and} \quad \frac{1}{a} = a^{-1},$$

where a is any nonzero real number



3.4.5 Guided Practice: Rewriting Negative Exponents

Rewrite each expression as an equivalent exponential expression using positive exponents.

1. $8^{-3} =$ _____

2. $\frac{1}{-4^{-5}} =$ _____

3. $(-2)^{-8} =$ _____

4. $\frac{1}{10^{-2}} =$ _____

5. $-5 \cdot 6^{-4} =$ _____

6. $\frac{2^{-3}}{9^{-2}} =$ _____



3.4.6 Collaboration: Applying Other Exponent Laws With Negative Exponents

Work with your partner to complete the following.

- Consider the exponential expressions in the table.
 - For each expression, work with your partner to rewrite the exponential expression in expanded form and evaluate. The first row has been partially completed.

Exponential Expression	Apply Negative Exponent Law	Expanded Form	Value
$6^{-2} \cdot 6^5$	$\frac{1}{6^2} \cdot 6^5$	$\frac{1}{6 \cdot 6} \cdot 6 \cdot 6 \cdot 6 \cdot 6 \cdot 6 =$ $\frac{1}{6} \cdot \frac{1}{6} \cdot 6 \cdot 6 \cdot 6 \cdot 6 \cdot 6$	
$(-4)^3 \cdot (-4)^{-7}$			
$3^{-4} \cdot 3^4$			

- b. Complete the table to evaluate each expression by applying the product of powers law.

Exponential Expression	Apply Product of Powers Exponent Law	Value
$6^{-2} \cdot 6^5$	$6^{-2+5} = 6^{\boxed{}}$	
$(-4)^3 \cdot (-4)^{-7}$		
$3^{-4} \cdot 3^4$		

- c. With your partner, discuss your observations of the tables from Parts A and B.

2. Consider the exponential expressions in the table.

- a. For each expression, work with your partner to rewrite the exponential expression in expanded form and evaluate. The first row has been partially completed.

Exponential Expression	Apply Negative Exponent Law	Expanded Form	Value
$(-8)^{-4} \div (-8)^{-6}$	$\frac{1}{(-8)^4} \cdot (-8)^6$	$\frac{(-8)(-8)(-8)(-8)(-8)(-8)}{(-8)(-8)(-8)(-8)}$	
$7^0 \div 7^{-4}$			
$9^{-5} \div 9^3$			

- b. Complete the table to evaluate each expression by applying quotient of powers law.

Exponential Expression	Apply Quotient of Powers Exponent Law	Value
$(-8)^{-4} \div (-8)^{-6}$	$(-8)^{-4-(-6)} = (-8)^{\boxed{}}$	
$7^0 \div 7^{-4}$		
$9^{-5} \div 9^3$		

- c. With your partner, discuss your observations of the tables after completing Parts A and B.

3. Several equivalent expressions are shown using the negative exponent law.

$$(5^{-3})^2 = \left(\frac{1}{5^3}\right)^2 = \left(\frac{1}{5 \cdot 5 \cdot 5} \cdot \frac{1}{5 \cdot 5 \cdot 5}\right) = \frac{1}{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5} = 5^{-6}$$

Which exponent law(s) can also be used to show $(5^{-3})^2 = 5^{-6}$?

4. Complete the statement.

The laws of exponents

- ☐ do
☐ do not

hold true for

numeric expressions with rational bases and negative exponents.



3.4.7 Wrap-Up: Cool Down

1. Ellen says that the value of 5^{-2} is -25. Explain why she is incorrect. Show the work to arrive at the correct answer.